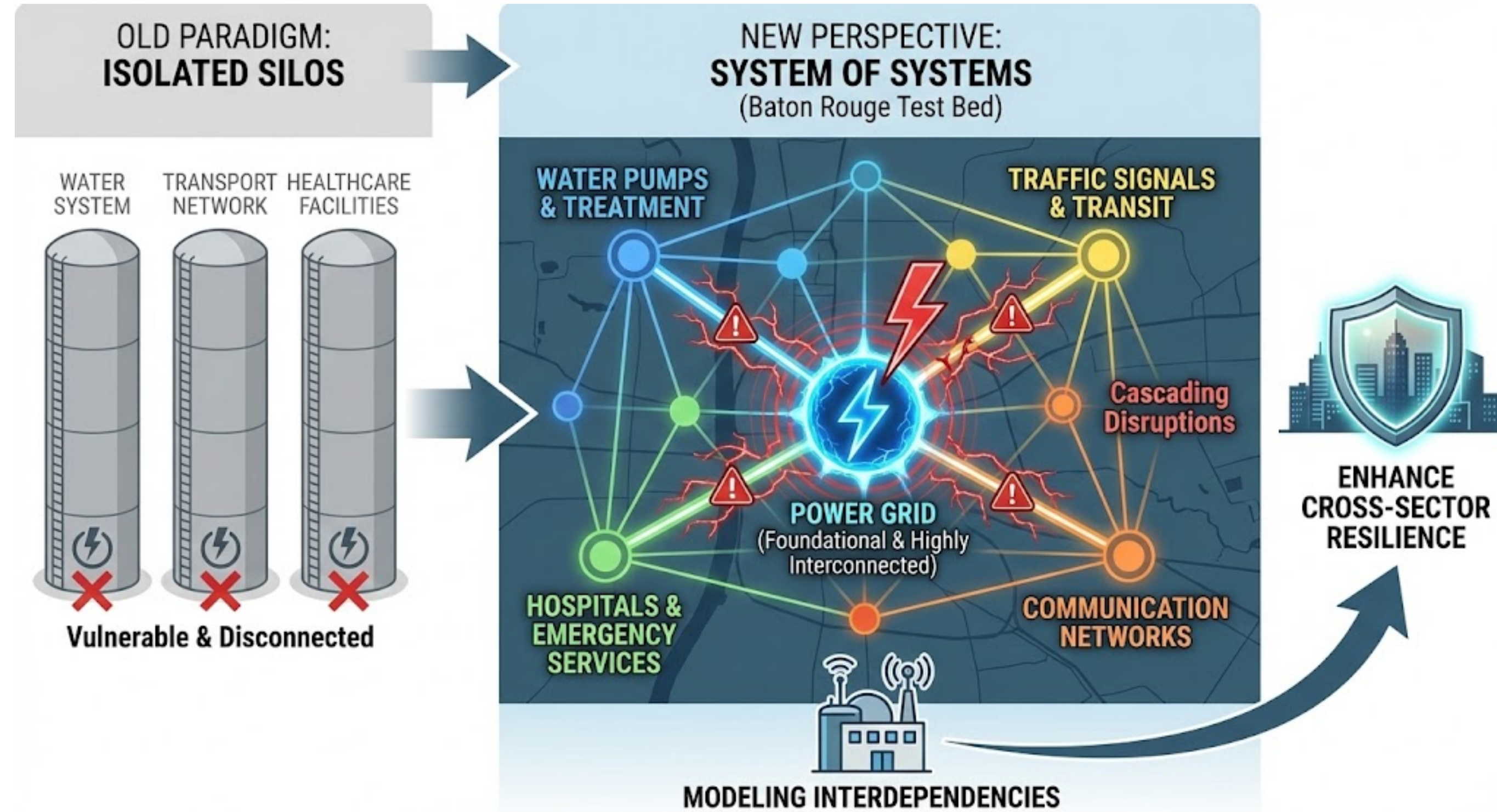
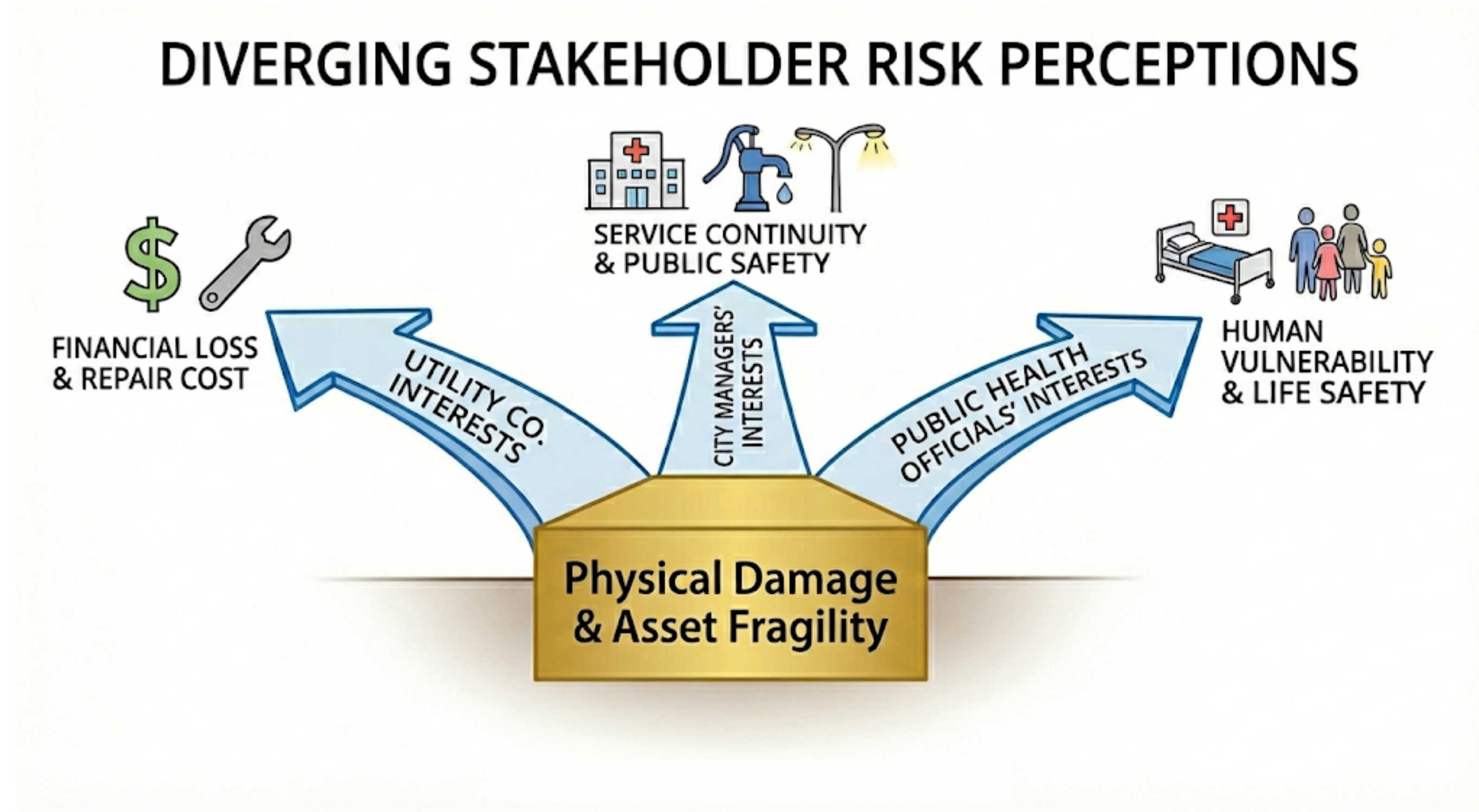


Overarching Objective:

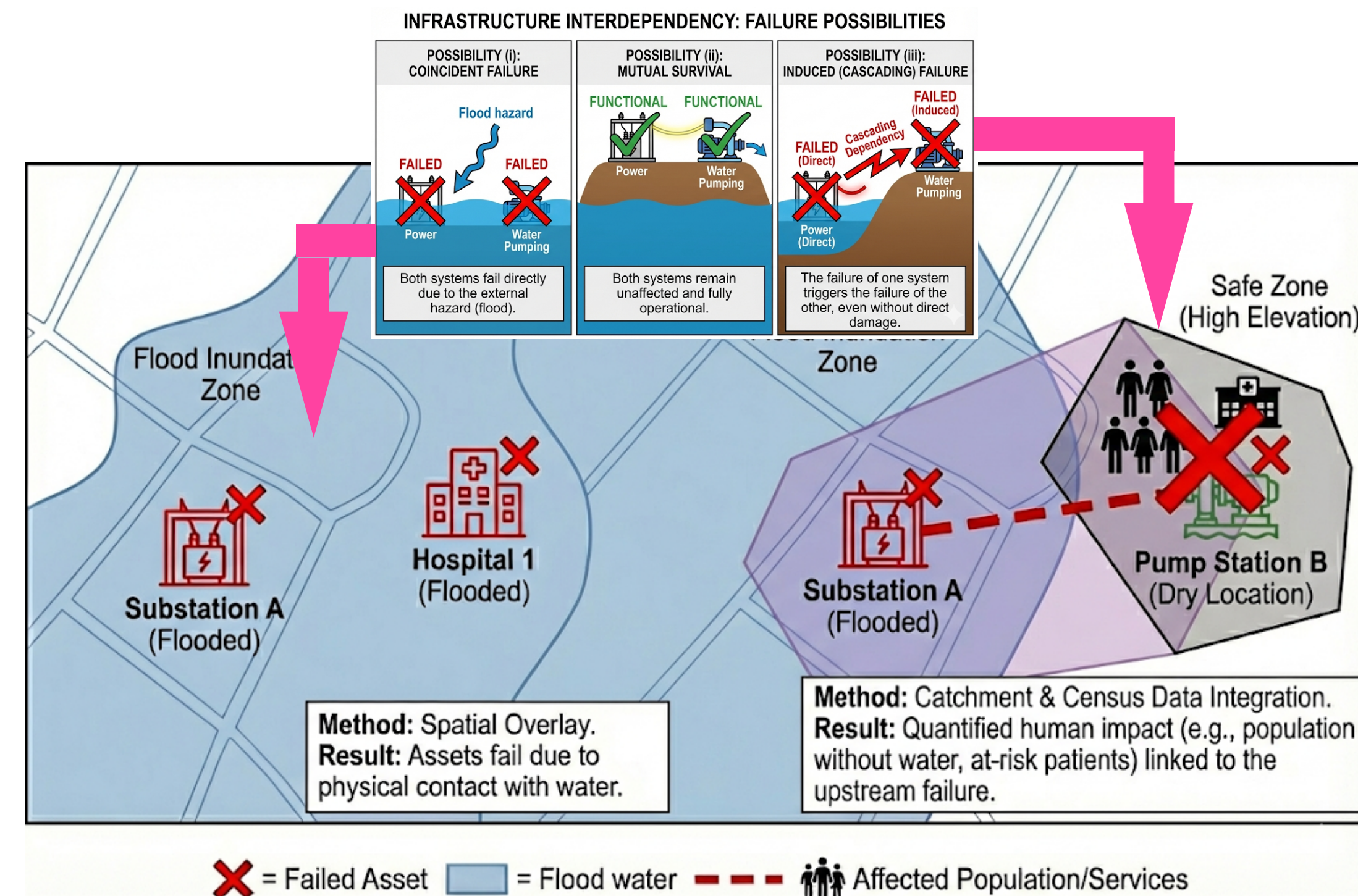


Modern urban infrastructure can no longer be viewed through the lens of isolated silos. Effectively managing these assets requires a '**system of systems**' perspective, acknowledging that a single failure can trigger **cascading disruptions** due to deep-rooted interdependencies. The power grid, as a foundational and highly interconnected element of this framework, perhaps exerts the largest influence on other systems. Our objective is to establish a **test bed in Baton Rouge** to rigorously model these interdependency effects and help create pathways to enhance cross-sector resilience.

Pressing Demand: Damage & Risk Diagnostics



Risk, and the tolerance for it, is inherently subjective. Stakeholders often possess diverging perceptions of risk driven by their unique priorities, institutional interests, and resource constraints. Consequently, a single, universal risk metric is often insufficient; a **multi-stakeholder risk quantification (MSRQ)** framework provides a more pragmatic and comprehensive approach. Regardless of the specific stakeholder perspective, **the foundational starting point** for any such quantification remains the **fragility of the assets and the system functionality**.



To evaluate urban infrastructure interdependency, we consider three failure states under flood forcing: (i) coincident failure of both systems, (ii) mutual survival, and (iii) indirect failure, where the disruption of one system triggers the collapse of another. This third state—**indirect failure**—is the primary indicator of systemic interdependency. By mapping the specific municipal assets and populations served by each power substation, we can quantify the **Value at Risk (VAR)** for every node in the grid. This approach identifies 'linchpin' substations whose criticality is defined not by their physical value, but by the vital services and socioeconomic dependencies they sustain.

Our progress:

Data Collection & ARC-GIS Integration

1. Power Grid Infrastructure (Baton Rouge)

- **Substation Profiles:** Geospatial coordinates, base elevations, and generation capacities (MWh/fuel source) for Entergy-owned distribution substations.
- **Ancillary Assets:** Geospatial mapping of Entergy-owned streetlighting networks.
- **Vulnerability Functions:** Preliminary depth-damage functions, including repair costs and recovery timeframes indexed to varying inundation depths.

2. Critical Municipal Infrastructure

- **Public Works & Health:** Geospatial coordinates and elevations for stormwater pumping stations, wastewater treatment plants, and healthcare facilities (hospitals).
- **Service Dependencies:** Mapping of streetlight distribution to assess localized visibility and safety impacts during outages.

3. Flood Hazard & Risk Assessment

- **Storm Surge Modeling:** SLOSH-based categorical inundation maps for Category 1 through 5 hurricanes, including depth increments from 3 ft to 9 ft.
- **Regulatory Risk:** FEMA National Flood Hazard Layer (NFHL) annual inundation risk maps.
- **Climatological Context:** Historical hurricane tracks and associated peak surge data.

4. Socioeconomic Data & Catchment Analysis

- **Substation Catchment Areas:** Delineated service areas for each substation, derived from historical outage maps.
- **Critical Load Estimation:** Mapping of hospitals and municipal facilities within respective catchment areas, including AI-estimated power consumption profiles and hospital bed capacities.
- **Demographic Exposure:** Census tract data integrated to estimate population density and vulnerability metrics per grid node.

Stakeholder	Primary Risk Interest	Key Data Layers Used	Critical Metric (Output)
Utility (Entergy)	Financial & Asset Integrity	Substation elevations, generation capacity, repair cost/time data, flood maps (SLOSH/FEMA).	Expected Annual Loss (EAL): Estimated CAPEX for repairs and lost revenue from unserved MWh.
City/ Municipal Managers	Service Continuity & Public Safety	Pumping station & sewage plant locations, hospital coordinates, streetlight networks, flood maps.	Service Outage Duration: Probability and duration of critical service failure (e.g., "Pumps-off" hours).
Public Health Officials	Human Vulnerability & Life Safety	Hospital bed capacity, AI-estimated power consumption, Census tract population data.	Patient-at-Risk Index: Number of compromised hospital beds or medically vulnerable populations per node.
Emergency Responders	Operational Accessibility	Historical hurricane tracks, streetlighting layers, catchment areas, inundation depths.	Response Impediment Score: Identification of "dark zones" and flooded access routes during a surge.